

SCH4U DIY Thesis Lab: Pre-Lab

A Non-Standard Cu/Zn Vinegar Cell and the Nernst Equation

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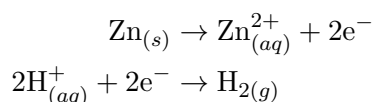
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Purpose

To compare the measured voltage of non-standard vinegar cells using copper as the inert cathode with the theoretical cell potential predicted by standard electrode potentials referenced to the SHE.

Background

The experimental cell uses an inert copper strip as the cathode surface, while the actual reduction is hydrogen ion reduction:



The standard hydrogen electrode (SHE) potential is defined as 0.00 V. Using standard reduction potentials gives the theoretical reference values:

- Zn/Zn²⁺ vs SHE: $E_{cell}^{\circ} = +0.76 \text{ V}$
- Al/Al³⁺ vs SHE: $E_{cell}^{\circ} = +1.66 \text{ V}$

Because vinegar is a weak acetic acid solution, the actual hydrogen ion concentration is much lower than 1.0 M. This means the measured voltage will be lower than the standard SHE-based values, and the Nernst equation can be used to explain the difference.

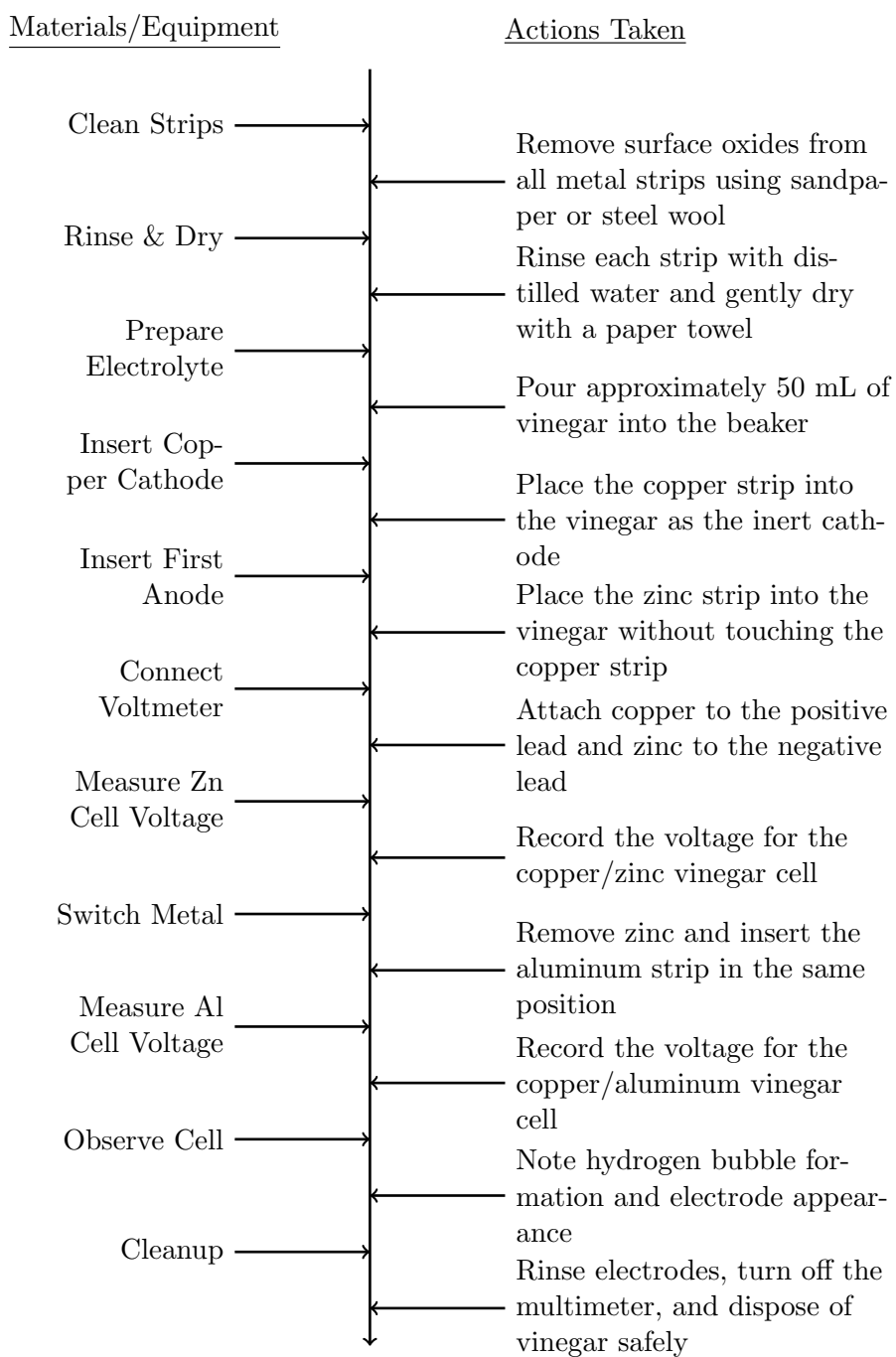
The Nernst equation is written as:

$$E_{cell} = E_{cell}^{\circ} - \frac{0.0592 \text{ V}}{n} \log(Q)$$

Materials

- 100 mL beaker
- Copper strip (electrode)
- Zinc strip (electrode)
- White vinegar (acetic acid solution)
- Multimeter with leads
- Alligator clips
- Distilled water (for rinsing)
- Sandpaper or steel wool (for cleaning metal strips)
- Safety goggles and gloves

Procedure Flowchart



Observation Table

Cell Combination	Voltage (V)	Observations
Copper / Zinc		
Copper / Aluminum		

Safety and Cleanup

- Wear safety goggles and gloves at all times.
- Avoid direct contact with vinegar on skin and clothing.
- Keep electrodes separated so they do not short circuit.
- Rinse electrodes with distilled water after the experiment.
- Dispose of the used vinegar according to your teacher's instructions.